First-Step Analysis: 10 Hard Problems

Expected Times · A-before-B Probabilities · Fundamental Matrix · Capstone

Stage-1 first-step questions disguise a tiny linear system inside a story — a coin pattern, a cube, a frog, a tennis game. The hard part is choosing the *right state space* (suffixes, distances, positions) so that only a handful of equations remain. Every option set has one correct answer and three answers built from a specific, common error; learn to recognize your own mistake in the wrong options.

I. Expected Hitting Times

Problem H1 — Two patterns racing: HHT or THH

A fair coin is flipped until *either* HHT or THH appears. What is the expected number of flips?

Options: (A) 4.0 (B) 5.0 (C) 6.5 (D) 8.0

Solution. Use the longest-useful-suffix states $\varnothing$, H, T, HH, TH. Let e_s be the expected remaining flips:

$$e_{HH} = 1 + \tfrac{1}{2}e_{HH} \;\Rightarrow\; e_{HH} = 2$$

(an H keeps you at HH, a T ends the game).

$$e_{TH} = 1 + \tfrac{1}{2}e_T, \qquad e_T = 1 + \tfrac{1}{2}e_T + \tfrac{1}{2}e_{TH}.$$

Substituting: $e_T = 1.5 + 0.75e_T \Rightarrow e_T = 6$, $e_{TH} = 4$. Then

$$e_H = 1 + \tfrac{1}{2}e_{HH} + \tfrac{1}{2}e_T = 5, \qquad e_\varnothing = 1 + \tfrac{1}{2}e_H + \tfrac{1}{2}e_T = \boxed{6.5}.$$

Trap analysis: (D) is the waiting time for HHT *alone* (8): with two targets the wait must be shorter. (A) halves 8, assuming "two patterns means half the time", but the patterns overlap heavily and do not split the flips evenly. (B) is e_H, the expected remaining time *after* a first head, not from the start.

Correct: (C) 6.5

★ *Intuition:* Two traps set in the same field: after HH only a T can end the game (and it ends it in HHT's favour), while after T the game is almost certainly decided in THH's favour (Penney: THH wins $\frac{3}{4}$ of the time). Having two ways to win shortens the wait, but not by a clean factor.

► *Speed insight:* Always sanity-check the answer against the one-pattern values: it must be *below* $\min(8, 8)$ and above the shortest possible finish (3).

Problem H2 — The cube's antipode

A walker moves along the edges of a cube, choosing one of the 3 edges at each vertex uniformly. Starting at a vertex, what is the expected time to reach the *opposite* vertex?

Options: (A) 6 (B) 8 (C) 10 (D) 12

Solution. By symmetry only the distance $d \in \{0, 1, 2, 3\}$ from the start matters. From $d = 0$ all 3 neighbours are at distance 1; from $d = 1$ one neighbour is back (prob. $\frac{1}{3}$) and two go forward ($\frac{2}{3}$); from $d = 2$ two go back ($\frac{2}{3}$) and one reaches the target ($\frac{1}{3}$):

$$k_0 = 1 + k_1, \quad k_1 = 1 + \tfrac{1}{3}k_0 + \tfrac{2}{3}k_2, \quad k_2 = 1 + \tfrac{2}{3}k_1.$$

Substituting gives $k_2 = 7$, $k_1 = 9$, and

$$k_0 = \boxed{10}.$$

Trap analysis: (B) is the number of vertices (8): a cover-style guess. (D) is the number of edges (12). (A) is 2·3, an unjustified "dimension times 2" heuristic. The correct 10 comes only from solving the lumped system.

Correct: (C) 10

★ *Intuition:* The walk is a tug-of-war on a 3-step ladder: the farther you are from the start, the stronger the pull *back* ($\frac{1}{3} \to \frac{2}{3}$). The expected hitting time of the antipode on the n-cube is $\sim 2^n$; for $n = 3$ it is 10.

► *Speed insight:* Lump symmetric states by "distance to the target." Four states instead of eight, and the equations are one-liners.

Problem H3 — Cover time of a complete graph

A walker moves to a uniformly random *other* vertex on the complete graph K_4. Starting at one vertex, what is the expected number of steps until all 4 vertices have been visited?

Options: (A) 3.0 (B) 4.5 (C) 5.5 (D) 6.0

Solution. Let j be the number of distinct vertices visited so far. At any visited vertex all 3 other vertices are neighbours, of which $4 - j$ are new, so the wait for the next new vertex is geometric with success probability $\frac{4-j}{3}$:

$$\mathbb{E}[T] = \underbrace{1}_{j=1} + \underbrace{\tfrac{3}{2}}_{j=2} + \underbrace{3}_{j=3} = \boxed{5.5}.$$

Trap analysis: (A) is the minimum possible (3 steps): every step visits a new vertex, which almost never happens. (B) $= 1.5 + 3$ forgets the certain first step. (D) $= 1 + 2 + 3$ uses success probabilities $\frac{1}{4-j} \to \frac{4-j}{4}$, i.e. forgets that the walker *cannot stay* at its current vertex.

Correct: (C) 5.5

★ *Intuition:* This is the coupon-collector problem with a twist: you can never "redraw" the coupon you hold. Early coupons are free and the last one is the expensive one — exactly the cheap-start, costly-finish shape of Chapter 1.4's harmonic sums.

► *Speed insight:* Cover times on symmetric graphs reduce to $\sum$ of geometric waits: for each stage, (success probability) = (number of new neighbours)/(degree).

Problem H4 — A sticky wall

A symmetric walk on $\{0, 1, \ldots, 6\}$ moves ± 1 with probability $\frac{1}{2}$ from states $1, \ldots, 5$; at 0 it *stays* with probability $\frac{1}{2}$ and moves to 1 with probability $\frac{1}{2}$; state 6 is absorbing. Starting from 0, find the expected number of steps to reach 6.

Options: (A) 30 (B) 36 (C) 42 (D) 49

Solution. Let k_i be the expected time from i and $g_i = k_i - k_{i+1}$. At 0: $k_0 = 1 + \frac{1}{2}k_0 + \frac{1}{2}k_1 \Rightarrow g_0 = 2$. At $i \geq 1$: $k_i = 1 + \frac{1}{2}k_{i-1} + \frac{1}{2}k_{i+1} \Rightarrow g_i = 2 + g_{i-1}$. Hence $g_i = 2(i + 1)$ and

$$k_0 = \sum_{i=0}^{5} g_i = 2(1 + 2 + \cdots + 6) = \boxed{42} = N(N + 1).$$

Trap analysis: (B) $36 = N^2$ is the answer for a *hard* reflecting wall (from 0 move to 1 with probability 1): $g_i = 2i + 1$. The sticky wall's self-loop wastes extra time. (A) $= N(N - 1)$ and (D) $= (N + 1)^2$ are other tempting polynomial guesses without derivation.

Correct: (C) 42

★ *Intuition:* A wall that sometimes says "try again" is a speed bump: each time you are at 0 you waste a step with probability $\frac{1}{2}$, and you visit 0 many times on the way. The extra cost is exactly $N = 6$ extra steps.

► *Speed insight:* The difference $g_i = k_i - k_{i+1}$ turns every "reflecting wall" problem into an arithmetic progression. Hard wall: $g_i = 1 + 2i$ (sum N^2). Sticky wall: $g_i = 2 + 2i$ (sum $N(N + 1)$).

II. Probability of Reaching A Before B

Problem H5 — Bold play beats timid play (Mosteller, Problem 37)

A gambler wins each even-money bet with probability $p = 0.45$. Starting with \$5 he wants to reach \$20. *Bold play* bets $\min(x,\ 20 - x)$ each time. What is his probability of reaching \$20?

Options: (A) 0.0318 (B) 0.2025 (C) 0.2500 (D) 0.4500

Solution. The reachable fortunes are $\{0, 5, 10, 15, 20\}$:

$$h_{10} = p = 0.45 \quad \text{(bet all \$10)},$$

$$h_5 = p\, h_{10} = 0.45(0.45) = \boxed{0.2025}.$$

(Check: $h_{15} = p + q\, h_{10} = 0.6975$.) Betting \$1 at a time ("timid play") would give $\frac{1-r^5}{1-r^{20}}$ with $r = \frac{11}{9}$, only ≈ 0.0318.

Trap analysis: (A) is the timid-play probability. (C) is $\frac{5}{20}$, the answer for a *fair* game. (D) is a single bet's win probability, ignoring that two successive wins are needed.

Correct: (B) 0.2025

★ *Intuition:* In an unfavourable game, every extra bet is another chance for the house edge to bite, so you should place as *few* bets as possible. This is Mosteller's Problem 37 ("bold play vs. cautious play"): in the book \$20 → \$40 gives ≈ 0.474 boldly versus ≈ 0.11 timidly.

▶ *Speed insight:* In a losing game, minimize the number of bets (bold play). In a winning game, maximize it (timid play) — the reverse of intuition.

Problem H6 — A frog on stepping stones

A frog starts on stone 0 and each jump moves it $+1$ or $+2$ stones with probability $\frac{1}{2}$ each. What is the probability that it lands on stone 5?

Options: (A) 0.500 (B) 0.625 (C) 0.656 (D) 0.667

Solution. Let $u_n = P(\text{land on } n)$. Condition on the *last* jump into n: it came from $n-1$ or $n-2$:

$$u_n = \tfrac{1}{2}u_{n-1} + \tfrac{1}{2}u_{n-2}, \qquad u_0 = 1,\ u_1 = \tfrac{1}{2}.$$

$$u_2 = 0.75, \quad u_3 = 0.625, \quad u_4 = 0.6875, \quad u_5 = \boxed{\tfrac{21}{32} = 0.65625}.$$

Trap analysis: (D) is the limit $\frac{2}{3} = \frac{1}{\text{mean jump}} = \frac{1}{1.5}$, reached only asymptotically; the sequence overshoots and undershoots it. (B) is u_3: an off-by-two indexing error. (A) assumes landing on any stone is a coin flip.

Correct: (C) 0.656

★ *Intuition:* The frog lands on a given stone about two-thirds of the time because its average jump of 1.5 stones means it "touches" on average one stone in every 1.5. The first few stones oscillate around $\frac{2}{3}$ and converge quickly, because the memory shrinks by $\frac{1}{2}$ every step.

▶ *Speed insight:* First-step ("last-jump") recurrences $u_n = \sum_j p_j u_{n-j}$ are renewal equations; the limit is $1/\mu$ where μ is the mean jump size.

Problem H7 — A fair game with overshoot

A walker starts at 3 and each step is $+2$ with probability $\frac{1}{3}$ or -1 with probability $\frac{2}{3}$ (mean step 0). It stops when it reaches 0 or first reaches a value ≥ 6 (i.e. it may stop at 6 or 7). Find $P(\text{it stops at } 0)$.

Options: (A) 0.467 (B) 0.500 (C) 0.533 (D) 0.667

Solution. Let $h_i = P(\text{stop at } 0 \mid \text{start } i)$ for $i = 1, \ldots, 5$ with $h_0 = 1$, $h_6 = h_7 = 0$:

$$h_i = \tfrac{1}{3}h_{i+2} + \tfrac{2}{3}h_{i-1}.$$

Solving the 5×5 system (eliminate $h_5 = \frac{2}{3}h_4$, $h_4 = \frac{2}{3}h_3$, …) gives

$$h_3 = \frac{8}{15} \approx \boxed{0.533}.$$

Consistency check by optional stopping: the walk is a martingale, so $\mathbb{E}[X_T] = 3$. With absorption probabilities $(0,\ 6,\ 7) \leftrightarrow (\frac{8}{15}, \frac{4}{15}, \frac{3}{15})$, $6 \cdot \frac{4}{15} + 7 \cdot \frac{3}{15} = 3$ ✓.

Trap analysis: (B) is $1 - \frac{3}{6}$: the fair-game formula, which assumes you stop *exactly* on the upper barrier. Here the walk can overshoot to 7, which changes the mean of the stopping value. (D) is $\frac{2}{3}$, the probability of a single down-step, ignoring the structure. (A) is $1 - \frac{8}{15} = \frac{7}{15}$, the probability of stopping *above*.

Correct: (C) 0.533

★ *Intuition:* A fair game does not guarantee a fair split of outcomes if the walker can *leap over* the finish line. Overshoot lets some of the upper-barrier mass land on 7 instead of 6, which means the walker needs less probability on the upper side to keep the mean at 3: so it *stops more often at* 0 than the naive $\frac{1}{2}$.

▶ *Speed insight:* Martingale ⇒ $\mathbb{E}[X_T] = X_0$ is a valid constraint, but with overshoot there are *three* stopping values, so one constraint is not enough — you still need a first-step system.

III. The Fundamental Matrix

Problem H8 — Expected visits in a fair gambler's ruin

A fair walk on $\{0, 1, 2, 3, 4\}$ is absorbed at 0 and 4. Starting from 1, what is the *expected number of visits to state* 3 before absorption?

Options: (A) 0.333 (B) 0.500 (C) 1.000 (D) 1.500

Solution. With transient states $1, 2, 3$, Q has $\frac{1}{2}$ on the off-diagonals, and

$$N = (I - Q)^{-1} = \begin{pmatrix} 1.5 & 1 & 0.5 \\ 1 & 2 & 1 \\ 0.5 & 1 & 1.5 \end{pmatrix}.$$

N_{13} is the expected number of visits to 3 starting from 1:

$$N_{13} = \boxed{0.5}.$$

Cross-check: the *probability* of ever visiting 3 is $\frac{N_{13}}{N_{33}} = \frac{0.5}{1.5} = \frac{1}{3}$ (also the gambler's-ruin probability $\frac{1}{3}$ of reaching 3 before 0), and each visit is followed by an expected 1.5 visits in total.

Trap analysis: (A) is $\frac{1}{3}$, the probability of *ever* visiting 3, not the expected number of visits (a burst of visits follows each arrival). (D) is $N_{33} = 1.5$, the expected visits starting *from* 3. (C) is $N_{22}/2$ or N_{12}: the wrong entry.

Correct: (B) 0.500

★ *Intuition:* Think of visits to room j as a Poisson-ish burst: with probability $\frac{1}{3}$ you reach it at all, and once there you collect on average $N_{33} = 1.5$ visits. Multiplying gives $\frac{1}{3} \times 1.5 = 0.5$ — the entry N_{13}.

▶ *Speed insight:* $N_{ij} = P(i \text{ ever visits } j) \cdot N_{jj}$. Both factors are easy to get separately when the chain has a gambler's-ruin structure.

Problem H9 — A maze with three exits

Three rooms. Room 1: to Room 2 w.p. 0.4, Room 3 w.p. 0.2, exit A w.p. 0.4. Room 2: to Room 1 w.p. 0.3, Room 3 w.p. 0.3, exit B w.p. 0.4. Room 3: to Room 1 w.p. 0.2, Room 2 w.p. 0.2, exits A, B, C w.p. $0.1, 0.2, 0.3$. Starting in Room 1, find $P(\text{leave through } C)$.

Options: (A) 0.129 (B) 0.200 (C) 0.300 (D) 0.355
Solution. Only Room 3 touches exit C. Let $h_i = P(\text{exit through } C \mid \text{start } i)$:

$$h_1 = 0.4h_2 + 0.2h_3, \quad h_2 = 0.3h_1 + 0.3h_3,$$

$$h_3 = 0.2h_1 + 0.2h_2 + 0.3.$$

From the first two, $h_1 = \frac{4}{11}h_3$ and $h_2 = \frac{4.5}{11}h_3$. Substituting in the third:

$$h_3\left(1 - \tfrac{1.7}{11}\right) = 0.3 \Rightarrow h_3 = \tfrac{11}{31}, \qquad h_1 = \tfrac{4}{11}\cdot\tfrac{11}{31} = \boxed{\tfrac{4}{31} \approx 0.129}.$$

(Full exit distribution from Room 1: $A = \frac{17}{31}$, $B = \frac{10}{31}$, $C = \frac{4}{31}$.)
Trap analysis: (D) 0.355 $= \frac{11}{31}$ is the probability of exiting through C starting *in Room 3*. (C) is the one-step exit probability 0.3 from Room 3. (B) is the one-step probability 0.2 of moving from Room 1 to Room 3. None accounts for the chance of wandering among the rooms first.
Correct: (A) 0.129
★ *Intuition:* Exit C lives in the back room only, and the front rooms have their own, much easier exits (A and B). So most of the traffic leaves before it ever reaches the back room: from the front, the chance of using the back exit is about one in eight.
▶ *Speed insight:* When only one state touches the target, write the equation for that state first (it contains the "+0.3"), then express the others as multiples of it. This avoids inverting a 3×3 matrix.

IV. Capstone

Problem H10 — Winning a game of tennis from 0–0

A server wins each point independently with probability $p = 0.6$. A game is won by the first player to reach 4 points with a lead of at least 2 (deuce rules). What is the probability that the server wins the game?
Options: (A) 0.544 (B) 0.600 (C) 0.692 (D) 0.736
Solution. Three pieces. With $q = 0.4$: (i) Win without ever reaching deuce: $p^4(1 + 4q + 10q^2) = 0.1296(1 + 1.6 + 1.6) = 0.5443$. (ii) Reach deuce (3–3): $\binom{6}{3}p^3q^3 = 20(0.216)(0.064) = 0.2765$. (iii) From deuce the server wins with probability $\frac{p^2}{p^2+q^2} = \frac{0.36}{0.52} = 0.6923$.

$$P = 0.5443 + 0.2765 \times 0.6923 \approx \boxed{0.736}.$$

(The expected number of points played in the game is ≈ 6.48.)
Trap analysis: (A) 0.544 counts only games won *before deuce*, forgetting that the server can still win after reaching deuce. (C) 0.692 is the deuce-state value, not the whole-game value from 0–0. (B) 0.6 is the single-point probability: tennis's scoring *amplifies* a 60% point-winner into a 74% game-winner.
Correct: (D) 0.736
★ *Intuition:* Each layer of the scoring system (point, game, set, match) is a magnifier for the stronger player. A 60% edge on each point becomes a 74% edge in a game, and that edge keeps compounding through sets and matches.
▶ *Speed insight:* Decompose long absorbing chains into *(paths that avoid the hard core) + (prob. of reaching the core)×(core value)*. Here the core is deuce, whose value is the closed form $\frac{p^2}{p^2+q^2}$.

Calibration note: H2, H3, H5, H8 are realistic 90–150s Stage-1 questions once the state space is chosen well (distance, number visited, bold play, $N_{ij} = P \cdot N_{jj}$). H1, H4, H6, H7, H9, H10 need a pattern-suffix system, a difference-of-times trick, a renewal recurrence, an overshoot argument, or a decomposition around deuce — Stage-2 grade, but each uses only Chapter 4.2's first-step method.